

Mathematics

Stage 8

Paper 2

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square in brackets in mark scheme

When Square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0–30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

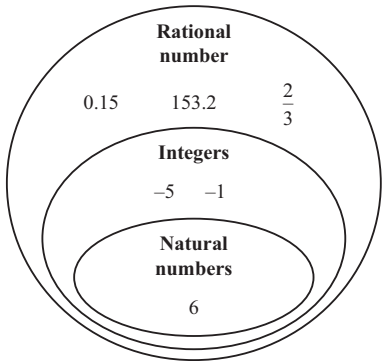
Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	$35m^2 + 14m$ final answer	1		
2	A correct explanation that the sample size is too small and/or may not be representative, e.g. he needs to sample more people.	1		Accept equivalent correct explanations that refer to sample size.
3	6300 correct answer only 0.73 correct answer only	2	Award 1 mark for each correct answer.	Answers must be in the correct order.
4	72	1		Accept more accurate value, e.g. 72.4205
5(a)	$x + 8 < 22$	1		Accept any clear indication.
5(b)	26	1		
6	8	1		
7	True False ☒ ☐ ☐ ☒ ☐ ☒ ☒ ☐	2	Award 1 mark for three correct answers.	Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance
8	$x - 3 = 20$ $10x + 2$ $A = \frac{1}{2} \times b \times h$ Formula Equation Expression	1		All three lines correct for the mark.
9	3.8749 and 3.865	1		Both answers correct for the mark. Accept any clear indication.
10	(x=) 21	1		
11	c a f	2	Award 1 mark for two correct answers.	All three answers correct for 2 marks. Answers must be in the correct order.

Question	Answer	Marks	Part Marks	Guidance
12	$5(3x - 5)$ final answer $13x(2x + 3)$ final answer	3	Award 1 mark for $5(3x - 5)$ and Award 2 marks for $13x(2x + 3)$ or Award 1 mark for $13(2x^2 + 3x)$ or $x(26x + 39)$	
13	$y = 3x - 7$ ☒ $y = 7x - 3$ ☐ $y = 7 - 3x$ ☐ $y = 7 + 3x$ ☒ $y = 3 - 7x$ ☐	1		Both ticks correct for the mark. Accept any clear indication.
14	$(a =) 23.5$	2	Award 1 mark for $[1269 \div] (\frac{1}{2} \times 12 \times 9)$ or better.	Or better, e.g. 54

Question	Answer	Marks	Part Marks	Guidance
15	2 : 3 : 5 correct answer only	3	Award 2 marks for any ratio equivalent to 2 : 3 : 5 e.g. $\frac{1}{5} : \frac{3}{10} : \frac{1}{2}$ or Award 1 mark for [dark] $\frac{1}{2}$ or equivalent.	Or equivalent, e.g. $\frac{5}{10}$
16(a)	interviews interviews questionnaires	1		All three answers in the correct order for the mark.
16(b)	A correct of explantion of why the sample size of boys and girls should not be equal, e.g. - There are more boys in the school so there should be more boys in the sample. - She has decided to sample the same proportion/10% of each gender.	1		Accept equivalent correct explanations.
16(c)	A correct explanation of the advantage of the sampling method, e.g. - This is a quick/convenient way to pick her sample. - She does not need to worry about the student not being in school.	1		Accept equivalent correct explanations.
17	Add 2 B (and sequence) C	2	Award 1 mark for each correct sentence If 0 marks scored, award 1 mark for two or three sequences correct out of [sequence B] 4, 10, 16, 22 [sequence C] 6, 12, 18, 24 [sequence D] 2, 4, 6, 8	Accept equivalents, e.g. + 2, plus 2 B and C can be in either order.

Question	Answer	Marks	Part Marks	Guidance
18		2	Award 1 mark for four or five numbers correctly placed.	
19	40	1		
20	$\begin{pmatrix} -4 \\ -2 \end{pmatrix}$	1		
21	$(h =) 600$ $(l =) 252\,000$	3	Award 2 marks for either of these: - one correct answer $h = 2^3 \times 3 \times 5^2$ and $l = 2^5 \times 3^2 \times 5^3 \times 7$ or Award 1 mark for any of these: $h = 2^3 \times 3 \times 5^2$ $l = 2^5 \times 3^2 \times 5^3 \times 7$ $h = 252\,000$ and $l = 600$	

Question	Answer	Marks	Part Marks	Guidance
22(a)	62 (seconds)	1		
22(b)	7	1		
22(c)	38	1		Accept any clear indication.
22(d)	A correct comparison and a correct median for group A, e.g. The median for group A is 79 seconds which means group A were faster on average (than group B).	2	Award 1 mark for correct median of 79 for group A or for correct comparison.	Do not accept comparisons without context, e.g. the median for group A is lower than the median for group B is not sufficient.
23	$4 - 0.6n$ or equivalent	2	Award 1 mark for sight of $[-]0.6n$ or for $4 + kn$ $k < 0$	
24	778 correct answer only	3	Award 2 marks for any of these: • 141 to 141.43 seen • An answer in the range 777 to 780.14...(that is not 778) • Correct full method $1.1 \times 100\,000 \div (\pi \times 22.5 \times 2)$ or equivalent or Award 1 mark for $\pi \times 22.5 \times 2$ or for <i>their</i> more accurate answer rounded correctly to the nearest whole number.	Accept values of π between 3.14 and $\frac{22}{7}$ For 2 marks, method may be seen in stages or in kilometres or metres.

Question	Answer	Marks	Part Marks	Guidance
25	Shows the total area is $7612.5 \text{ (m}^2\text{)}$ and correct supporting value, e.g. • [number of bags is] 13.2... • [13 bags cover] $7475 \text{ (m}^2\text{)}$ and correct conclusion, e.g. • 14 bags are required • 13 bags are not enough • $13.2... > 13$	4	Award 2 marks for 7612.5 or Award 1 mark for either of these: • one correct rectangle area from 5250, 1050, 2550, 3750, 8925 • $\frac{1}{2} \times (85 - 50) \times (105 - 30)$ or better. and Award 1 mark for either of these: • sight of 13.2... • sight of [13 bags covers] 7475 • <i>any area</i> $\div (115 \times \frac{25}{5})$ or better.	Implied by [both rectangle areas] 6300 Or better, e.g. $\frac{1}{2} \times 35 \times 75$, 1312.5 Or better, e.g. <i>any area</i> $\div 575$

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